# Audio and Signal Processing

2026311323 Howon Chung

## 1. Problem Statement

Given $X(\omega) = |X(\omega)|\, e^{j\angle X(\omega)}$, so that $\log X(\omega) = \log|X(\omega)| + j\angle X(\omega)$: prove that for a minimum-phase system, the phase $\angle X(\omega)$ is fully determined by the log-magnitude $\log|X(\omega)|$.

A minimum-phase sequence $x[n]$ is real, causal, and stable, with a causal and stable inverse $1/X(z)$ -- equivalently, every zero AND every pole of $X(z)$ lies strictly inside the unit circle.

## 2. Setup: the Complex Cepstrum

Define the complex cepstrum as the inverse transform of $\log X$: $\hat{x}[n] = \mathrm{IDFT}\{\log X(k)\}$, where $\log X(k) = \log|X(k)| + j\angle X(k)$. The proof proceeds in two steps: (1) show that the minimum-phase condition forces $\hat{x}[n]$ to be causal ($\hat{x}[n] = 0$ for $n < 0$); (2) show that for ANY real, causal sequence, the real and imaginary parts of its transform are a Hilbert-transform pair. Combining the two: $\log|X(\omega)|$ and $\angle X(\omega)$ are a Hilbert-transform pair whenever $x[n]$ is minimum phase, which is exactly the sense in which one determines the other.

## 3. Step 1 — Minimum Phase ⟹ Causal Cepstrum

Write $X(z)$ in factored (zero/pole) form. For a minimum-phase system, every zero $a_k$ and every pole $c_k$ satisfies $|a_k| < 1$, $|c_k| < 1$ (strictly inside the unit circle):

$$X(z) = A \cdot \Pi_k (1 - a_k z^{-1}) / \Pi_k (1 - c_k z^{-1}),\ |a_k| < 1,\ |c_k| < 1 \text{ for all } k$$

Taking the log turns the product into a sum: $\log X(z) = \log A + \Sigma_k \log(1 - a_k z^{-1}) - \Sigma_k \log(1 - c_k z^{-1})$. Each term expands as a power series using $\log(1-x) = -\Sigma_{\ell=1}^{\infty} x^\ell/\ell$, valid for $|x|<1$, i.e. for $|z| > |a_k|$ — a region that includes the unit circle since $|a_k| < 1$:

$$\log(1 - a_k z^{-1}) = -\Sigma_{\ell=1}^{\infty} (a_k^\ell/\ell)\, z^{-\ell}$$

Every term on the right is a NEGATIVE power of $z$ ($z^{-\ell}$, $\ell \geq 1$) — there is no $z^{+\ell}$ term at all. (Had $a_k$ instead been OUTSIDE the unit circle, the same expansion, now valid for $|z| < |a_k|$, would have produced only POSITIVE powers of $z$, i.e. an anticausal contribution — this is the exact mechanism that breaks down outside the minimum-phase case, used again in Section 6.) Summing every zero and pole term:

$$\log X(z) = \log A + \Sigma_{n=1}^{\infty} \hat{x}[n]\, z^{-n},\ \hat{x}[n] = (\Sigma_k a_k^n - \Sigma_k c_k^n) / n \text{ for } n \geq 1$$

No negative-$n$ term appears anywhere in this sum. Therefore $\hat{x}[n] = 0$ for $n < 0$: the complex cepstrum of a minimum-phase sequence is causal. This is the same expansion Chapter 6 (p.12) derives for a single zero factor, generalized here to the full minimum-phase product (every factor, not just one).

## 4. Step 2 — A Causal Real Sequence's Transform Has Hilbert-Paired Real/Imaginary Parts

This step uses no properties of $X$ specifically — it holds for any real, causal sequence $h[n]$ ($h[n]=0$ for $n<0$). Split $h[n]$ into even and odd parts: $h_e[n] = (h[n]+h[-n])/2$, $h_o[n] = (h[n]-h[-n])/2$. Because $h$ is causal:

$$n > 0:\ h[-n] = 0 \Longrightarrow h_e[n] = h_o[n] = h[n]/2$$
$$n < 0:\ h[n] = 0 \Longrightarrow h_e[n] = h[-n]/2 = -h_o[n]$$

Both cases combine into a single relation valid for every $n \neq 0$: $h_o[n] = \mathrm{sgn}(n) \cdot h_e[n]$. Taking the DTFT: $h_e[n]$ is even and real, so its transform $H_e(\omega)$ is real (and even); $h_o[n]$ is odd and real, so its transform $H_o(\omega)$ is purely imaginary (and odd). Hence $\mathrm{Re}\{H(\omega)\}$ comes entirely from $h_e$, and $\mathrm{Im}\{H(\omega)\}$ entirely from $h_o$. The time-domain relation $h_o[n] = \mathrm{sgn}(n) \cdot h_e[n]$ is precisely the operation that defines the discrete Hilbert transform in the frequency domain: multiplying a sequence by sgn(n) is the time-domain counterpart of taking its Hilbert transform. So for ANY real causal $h[n]$: $\mathrm{Im}\{H(\omega)\} = \mathrm{Hilbert}\{\mathrm{Re}\{H(\omega)\}\}$ (up to the fixed sign convention of the transform).

## 5. Combining the Two Steps

Apply Step 2 with $h[n] = \hat{x}[n]$. By Step 1, $\hat{x}[n]$ is causal whenever $x[n]$ is minimum phase, so Step 2's conclusion applies directly:

$$Re\{\log X(\omega)\} = \log|X(\omega)|,\ Im\{\log X(\omega)\} = \angle X(\omega)$$
$$\Longrightarrow \angle X(\omega) = Hilbert\{\log|X(\omega)|\}$$

This is the statement to prove: for a minimum-phase system, the phase is the (discrete) Hilbert transform of the log-magnitude — a fixed, invertible linear operation, not merely a loose correlation. Knowing $\log|X(\omega)|$ at every $\omega$ therefore determines $\angle X(\omega)$ completely (up to the n=0 cepstrum term $\hat{x}[0] = \log A$, an overall real gain that carries no phase information).

## 6. Why Minimum Phase Specifically — and the Reconstruction It Enables

Step 1 used BOTH directions of the zero/pole argument: a factor inside the unit circle contributes only to $n \geq 0$, and (by the same expansion, now centered outside the unit circle) a factor outside contributes only to $n < 0$. A general (mixed-phase) system has some zeros/poles outside, so its cepstrum has a genuine non-causal part, and Step 2 no longer applies — the log-magnitude and phase are then independent: many different phase responses are consistent with the same $|X(\omega)|$ (the differences are exactly the all-pass factors obtained by reflecting a zero from inside to outside the unit circle, which leave $|X(\omega)|$ unchanged but change $\angle X(\omega)$). Minimum phase is what removes this ambiguity.

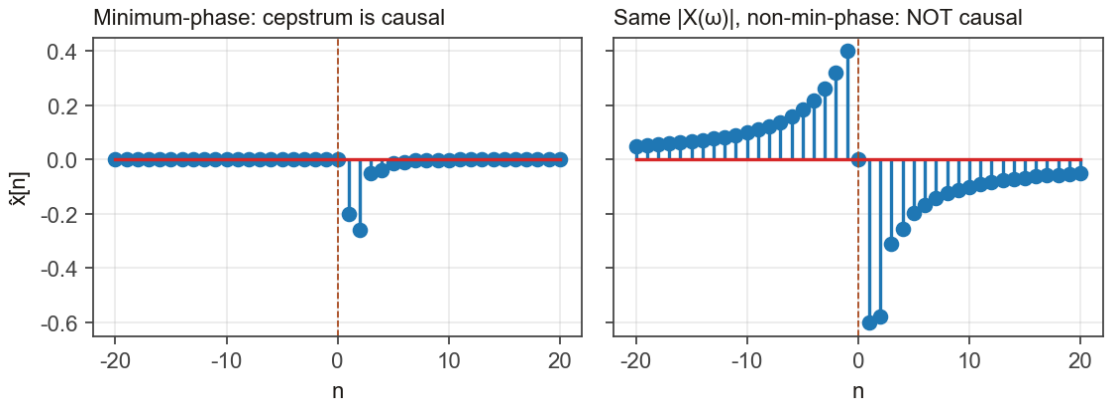
The proof is also constructive, matching the liftering procedure in Chapter 6 (p.31, "Zero- and Minimum-phase Synthesis"): from $\log|X(\omega)|$ alone, (i) inverse-transform to get the real cepstrum (automatically even, since $\log|X(\omega)|$ is real and even); (ii) keep the n=0 term, double every n>0 term, discard every n<0 term — this constructs a fully causal $\hat{x}[n]$ satisfying $\hat{x}_o[n]=\mathrm{sgn}(n)\hat{x}_e[n]$ exactly; (iii) forward-transform and exponentiate. The result's phase is not merely similar to the original -- it recovers the exact minimum-phase $\angle X(\omega)$, as verified numerically below.

## 7. Numerical Verification

Example: $X(z) = (1 - a_1 z^{-1})(1 - a_2 z^{-1})$ with $a_1$=0.6, $a_2$=-0.4 (both zeros strictly inside the unit circle — minimum phase). For contrast, a second system reflects $a_1$'s zero to $1/a_1$ (outside the unit circle) while keeping $a_2$ unchanged; by construction this has the IDENTICAL magnitude response (max |log-magnitude difference| = 7.8e-16) but is NOT minimum phase.

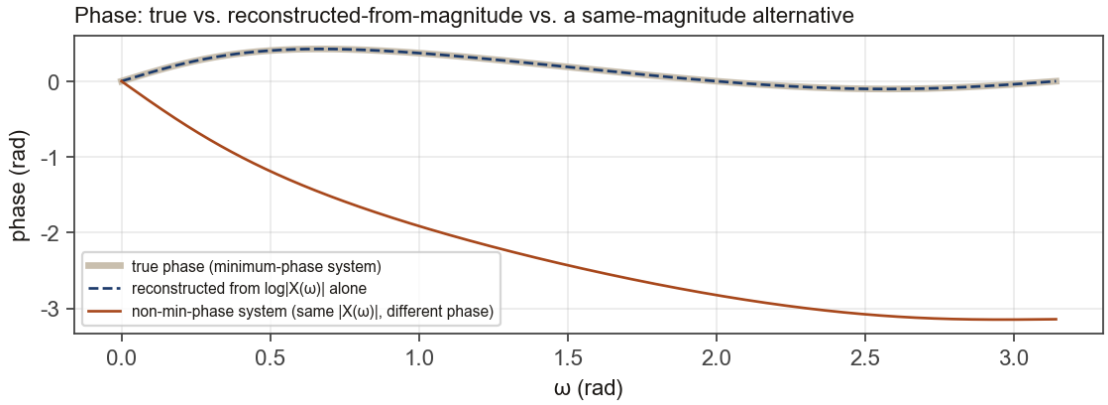
| Quantity | Minimum-phase system | Non-minimum-phase system (same $\|X(\omega)\|$) |
|---|---|---|
| max $\|\hat{x}[n]\|$ for n < 0 (should be 0 iff causal) | 7.70e-18 | 0.400 |

| | | |
|---|---|---|
| max $\lvert\hat{x}[n]\rvert$ for $n \geq 0$ | 0.260 | 0.600 |




The minimum-phase cepstrum's n<0 part is at machine-precision noise level (7.7e-18) against an n≥0 part of order 0.3 -- causal, as Section 3 predicts. The non-minimum-phase cepstrum's n<0 part (0.400) is the same ORDER OF MAGNITUDE as its n≥0 part -- genuinely non-causal, confirming the minimum-phase condition is what makes causality (and therefore the Hilbert-transform / determinacy result) hold.

Despite having identical magnitude, the two systems' phases differ by up to 3.142 rad (= π, exactly a half-cycle at ω=π in this example) -- concrete confirmation that log-magnitude alone does NOT determine phase once the minimum-phase restriction is dropped.




Reconstructing phase from the minimum-phase system's log-magnitude alone (Section 6's liftering procedure) matches the true phase to within 4.2e-16 rad, and the reconstructed log-magnitude matches to within 4.4e-16 -- both at floating-point noise level, i.e. an exact reconstruction up to numerical precision.

## 8. Conclusion

For a minimum-phase system, every zero and pole lies inside the unit circle, which forces the complex cepstrum to be causal (Section 3). Any real, causal sequence has real and imaginary transform parts related by the discrete Hilbert transform (Section 4) -- a general fact independent of this problem. Applying this to $\log X(\omega) = \log|X(\omega)| + j\angle X(\omega)$ gives $\angle X(\omega) = \text{Hilbert}\{\log|X(\omega)|\}$: the phase is the Hilbert transform of the log-magnitude, hence fully (and constructively) determined by it. The result fails outside the minimum-phase case because zeros/poles outside the unit circle contribute an anticausal part to the cepstrum, breaking the

causality Step 1 depends on -- confirmed numerically above by a same-magnitude, different-phase counterexample.